

Topological Feature Learning for Parameter Estimation in Neyman-Scott Processes

First-year progression report

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1 Topological feature learning for Neyman-Scott process parameters

1.1 Motivation and problem

Spatial point patterns – the locations of galaxies, cells, epicentres, or trees – appear across the sciences, and are commonly modelled as realizations of a spatial point process (?), whose parameters are themselves interpretable scientific quantities rather than curve-fitting artefacts. A Neyman-Scott process generates points in two stages: unobserved *parent* locations at intensity κ , each producing a Poisson-distributed number of *offspring* (mean μ) scattered around it according to a kernel with characteristic scale σ . When such a process models galaxy clustering, κ is the number density of haloes hosting galaxies, μ the mean galaxy occupancy of a halo, and σ the spatial extent of the galaxy distribution within a halo – exactly the quantities a halo occupancy model is built to constrain [TODO: cite halo occupancy reference]. In forest ecology, ? fit Thomas processes (a Neyman-Scott process with a Gaussian offspring kernel) to three tropical tree species and read the cluster scale σ directly as a summary of seed dispersal tightness, finding that species dispersed by exploding capsules, by wind, and by animals separate cleanly along that one parameter: recovering (κ, μ, σ) well is how the pattern gets turned back into an interpretable physical or ecological statement, not an end in itself.

We consider the setting a practitioner actually faces: a single observed point pattern, and one chance to estimate the parameters that generated it – not repeated draws of the same process. Existing estimators for this problem – minimum contrast fits a discrepancy between an empirical and theoretical summary curve (?), Palm likelihood builds a composite likelihood from the Palm intensity (?), fully Bayesian approaches sample the posterior over the latent parent locations by MCMC (?) – share three limitations that motivate this project. First, they need a closed-form summary statistic that may not exist at all (the LGCP-Strauss process of ? has an intractable density) or may exist but fail the regularity conditions the theory needs: ? require their K -function to be twice continuously differentiable in the parameters, which excludes the Matérn cluster process outright. Second, minimum contrast depends on hyperparameters (r_{max} , q , a power-transform exponent) chosen per dataset by rule of thumb, with no held-out criterion for whether the choice was good; ? report Thomas-process estimates of κ and μ that are “quite sensitive” to r_{max} and q , tracing this to a strong negative correlation between $\hat{\kappa}$ and $\hat{\mu}$ under which the product $\hat{\kappa}\hat{\mu}$ is stable while the individual estimates are not. Third, these methods become unstable near complete spatial randomness: for a Thomas process,

$$K(t) = \pi t^2 + \frac{1 - e^{-t^2/4\sigma^2}}{\kappa}, \quad (1)$$

so as σ grows at fixed t the second term vanishes and K tends to the Poisson curve *regardless of* κ (?). A discrepancy nearly flat along this trade-off yields estimates sensitive to the optimizer’s

starting point; ? report exactly this failure in their own simulations, weakly clustered patterns whose estimated K was indistinguishable from Poisson producing “rather extreme” clustering estimates. We deliberately retain this near-Poisson regime in our design distribution below, rather than restricting to the well-separated-cluster case classical methods are comfortable with.

These three limitations motivate features needing no closed form and no choice of summary statistic: a multi-scale, process-agnostic description of the point cloud, built directly from its geometry.

1.2 Approach

Persistent homology as a process-agnostic feature. A radially averaged statistic such as K or g collapses a point cloud to a single curve indexed by scale, discarding *how* density and gaps are distributed across space. Persistent homology instead tracks a growing family of simplicial complexes on the cloud as a scale r increases: at $r = 0$ every point is its own component, and as r grows, points connect, components merge, and loops open and close. Each such feature’s *birth* and *death* scale is recorded as a point (b, d) in a *persistence diagram* (points with $d - b$ small are usually treated as noise). The 0-dimensional diagram records when clusters merge; the 1-dimensional diagram records voids and gaps that persist across scales, capturing empty or repulsive structure a radially averaged statistic sees only indirectly. Neither needs a closed form or an assumption about the generating process family. A diagram is a variable-size multiset rather than a vector, so it must be *vectorized* before it can feed a neural network; we use *persistence images* (?), which smooth each diagram point with a Gaussian kernel and integrate onto a fixed pixel grid. Betti curves (?), persistence landscapes (?), and persistence silhouettes (?) are alternative vectorizations we compare against directly (Figure ??); none improved on persistence images under our architecture and design distribution (Section ??).

Simulator and design distribution. Training data are pairs $(\mathbf{x}_i, \theta_i)$, each a single realization: parents form a Poisson process on a slightly expanded window (so clusters whose parent lies just outside are not truncated), each parent independently produces a Poisson-distributed number of offspring, and each offspring is displaced from its parent by a draw from the kernel – Gaussian for Thomas, and, for nested Thomas, a second Thomas process standing in for the parent layer itself. Sampling θ_i uniformly over (κ, μ, σ) is inadequate because these processes separate into qualitative regimes occupying curved regions of that space, wasting most of the sampled mass on configurations that are near-Poisson or degenerate. We instead reparameterize into coordinates carrying the same meaning for every Neyman–Scott family member: the expected offspring intensity $\lambda = \kappa\mu$, the parent intensity κ , and a dimensionless overlap index

$$\nu = \sigma\sqrt{\kappa}, \quad (2)$$

the ratio of cluster radius to mean inter-parent spacing, and sample log-uniformly in (κ, λ, ν) subject to a point budget ($\lambda \in [150, 800]$, since persistence computation scales badly with cloud size), a resolvability constraint ($\kappa \in [15, 120]$), and an overlap range $\nu \in [0.10, 0.90]$ whose upper end deliberately keeps the near-Poisson regime of Section ?? in training rather than excluding it.

ParamNet. The model maps a cloud to a parameter estimate by

$$\text{cloud} \xrightarrow{\text{vectorize}} \text{persistence image} \xrightarrow{\text{encoder}} \text{embedding} \xrightarrow{\text{FCNN}} \hat{\theta}, \quad (3)$$

following the neural-estimator pattern introduced by ?; our contribution is the feature-extraction and calibration stage, its evaluation on a genuinely two-level process (nested Thomas), and a direct comparison of vectorizations. Images come from a density-sensitive DTM_k filtration rather than the more common Vietoris–Rips filtration: Rips uses raw inter-point distances and

vectorization_schematic.pdf

Figure 1: One point cloud $\rightarrow$ its persistence diagrams $\rightarrow$ the vectorizations compared: persistence-image raster, landscape stack, silhouette curve, Betti curve, and a persistence-statistics summary. Top: a tight-cluster design ($\kappa = 50, \mu = 4, \sigma = 0.01$); bottom: a near-Poisson design ($\kappa = 4, \mu = 5, \sigma = 0.5$) – the regime Section ?? argues classical methods handle badly.

is sensitive only to the sparsest connection between two regions, whereas DTM_k uses each point’s average distance to its k nearest neighbours and so responds directly to local density – exactly what a Neyman–Scott process’s parameters govern (?). The Thomas model uses a single scale ($k = 5$); nested Thomas fuses three scales ($k \in \{5, 10, 15\}$) through one shared convolutional encoder, which works better than giving each scale its own encoder (Section ??). Before rasterizing, each diagram’s axes are calibrated to a shared range fit on the training diagrams only – a per-diagram coverage-quantile rule, so one outlier diagram cannot compress every other diagram’s range toward the origin – then frozen before touching validation or test data. The encoder is a small CoordConv (?) convolutional stack: CoordConv appends each pixel’s normalized coordinates as extra input channels, giving the otherwise translation-equivariant convolutions access to absolute position, which matters because a persistence image’s axes are not arbitrary – birth encodes local density, persistence encodes separation – so a pixel’s location is informative in its own right. The resulting embedding is concatenated

Method	Thomas $\mathcal{L}$	Nested-Thomas $\mathcal{L}$
ParamNet	0.125 ± 0.007	0.192 ± 0.010
vihrs	0.115 ± 0.007	0.217 ± 0.006
Minimum contrast, K	0.855 ± 0.682	— ^a
Minimum contrast, g	6.199 ± 0.339	— ^a
Chance level (label shuffle)	1.003 ± 0.019	1.006 ± 0.020

Table 1: Mean squared error in log-normalized parameter space, $n = 10$ seeds, shared test split per process. ^aA closed-form K/g for nested Thomas was derived and fitted, but the derivation has not yet been independently validated, so we do not report the resulting numbers as a finding.

with $\log N(\mathbf{x})$ and passed to a small feed-forward head producing $\hat{\theta}$. The model is trained by minimizing mean squared error between $\hat{\theta}$ and θ in log-normalized parameter space (each parameter log-transformed and z-scored on the training split), so parameters spanning different orders of magnitude contribute comparably to the loss. Data are split by parameter vector, so no θ value appears in both training and evaluation, and error is reported in those same log-normalized units.

1.3 Results

We compare ParamNet against minimum contrast fit separately to K and to g (?), and against vihrs, our re-implementation of the neural estimator of ?: the same pipeline with the persistence-image feature replaced by a 1-D CNN over the classical, edge-corrected $L(r) - r$ curve plus $\log N$. A label-shuffle control – the same model retrained with training labels permuted – lands at $\mathcal{L} \approx 1.0$ on both processes and serves as a chance-level ceiling in Table ??.

Persistence images carry real signal about Neyman–Scott parameters: both processes land roughly an order of magnitude below the label-shuffle ceiling, and that signal sits in the images rather than in the point count – training on $\log N$ alone collapses to the same ≈ 0.9 – 1.0 chance level as label-shuffling, while removing $\log N$ from the full model costs little on nested Thomas ($0.192 \rightarrow 0.203$) and is merely noisy on Thomas. We report the comparison honestly rather than as an unqualified win, though: vihrs beats ParamNet on every Thomas parameter ($\mathcal{L}^K = 0.192$ vs. 0.201 , $\mathcal{L}^\mu = 0.122$ vs. 0.134 , $\mathcal{L}^\sigma = 0.029$ vs. 0.040), so a hand-built radial summary beats our topological pipeline outright on the single-level process. On nested Thomas the picture inverts only partially: ParamNet gains an advantage on the density and count targets (parent intensity, meta-offspring, mean offspring), while vihrs keeps its edge on both scale parameters – $L(r) - r$ is built to resolve a radius directly, a density-based topological feature is not, and it is only on the more complex two-level process that the trade-off favours our pipeline. A second, structural pattern holds regardless of estimator or process: parent intensity κ is the hardest target by a wide margin and cluster scale the easiest, reading as a property of the estimation problem rather than of any one feature set.

Both neural methods dominate minimum contrast on Thomas by roughly an order of magnitude, but for different reasons. On K , six of ten seeds land at 0.365 – 0.399 (close to vihrs) while the other four land at 0.903 – 2.056 – seed-to-seed instability in the near-Poisson regime of Section ??, not a uniformly bad fit. On g , every seed is worse than chance, pointing to a systematic bias rather than scattered non-convergence: our contrast exponent ($c = 1$, chosen by analogy with the K -based rule of thumb) was never tuned against the $c = 1/4$ value validated for aggregated patterns (?); we do not read it as a general verdict on contrast estimation. The same comparison was attempted for nested Thomas this month, against a closed-form K/g derived for the two-level process, but the derivation has not been checked independently and the resulting fits are far outside a plausible range, so we treat this baseline as still outstanding rather than report numbers.

We also compared three alternative vectorizations (Betti curves, landscapes, silhouettes) against persistence images under an identical encoder and design distribution; none improved on either process. Fusing the three DTM scales through one shared encoder likewise beats giving each scale its own encoder (0.192 vs. 0.222 nested-Thomas $\mathcal{L}$), which is why ParamNet shares an encoder rather than using independent branches.

1.4 Limitations and future directions

Limitations.

- **No advantage on the single-level process.** `vihrs` wins on every Thomas parameter and on both scale parameters of nested Thomas; our advantage is confined to the two-level process's density and count targets, and to three alternative vectorizations under one architecture and design distribution – evidence about this setting, not a general claim.
- **Multiple models, and expensive ones.** The number of filtration scales fused is a per-process choice made by comparing validation losses, undercutting the process-agnostic framing behind this work; always fusing three scales costs three times the persistence-computation cost for a Thomas change within seed noise, and persistence computation already scales badly with point count, which is why the design distribution caps the expected count at 800.
- **No irreducible error floor yet.** The natural lower reference for every number above – the dispersion of the best possible estimator across repeated realizations of the same θ – has been attempted but not obtained: the estimation script ran to completion but returned NaN throughout, traced to a suspected label-indexing bug not yet fixed.

Future directions.

- A universal, process-agnostic encoder, rather than a per-process filtration-scale choice; a small three-seed comparison also suggests a multiparameter persistence feature (a joint density–radius bifiltration fused with the `vihrs` branch) may close part of the Thomas gap, though not yet validated at the scale above.
- Inhomogeneous and non-stationary processes, where classical closed forms are least available and a topological approach has the most to gain: the natural comparator, the two-step procedure of ?, needs observed covariates and a correctly specified intensity, whereas a simulation-trained estimator needs neither.
- Real data: galaxy catalogues, motivated by Section ??, and the Barro Colorado Island tree data of ?, already fitted with a Thomas process.